

Marketplace Analyst

Round 1 submission for the Marketplace AI Competition. This proposal presents Marketplace Analyst as a grounded decision support layer for Marketplace style simulation performance and strategy.

This package goes a modest step beyond a text only answer: it already points to a live Vercel deployment you can click through, so the submission doubles as an early showcase of what the project looks like and how it behaves in a browser, not only as a written concept.

I am aware that Marketplace Simulations already ships an AI coach style experience inside its platform. What I am arguing here is that this design is a fuller and more helpful companion for the kind of work students do in the simulation: it is more visual, more flexible in how it presents evidence, and built to keep answers grounded in the quarter snapshot through citations and widgets, which can also make outcomes feel more accurate than a plain text chat alone for numeric, report heavy work.

Submission title: Marketplace Analyst: grounded decision support for simulation performance and strategy

Live demo: <https://marketplace-analyst.vercel.app>

Repository: <https://github.com/AlexGiurea/marketplace-analyst>

1. How are businesses currently using this idea in the workforce?

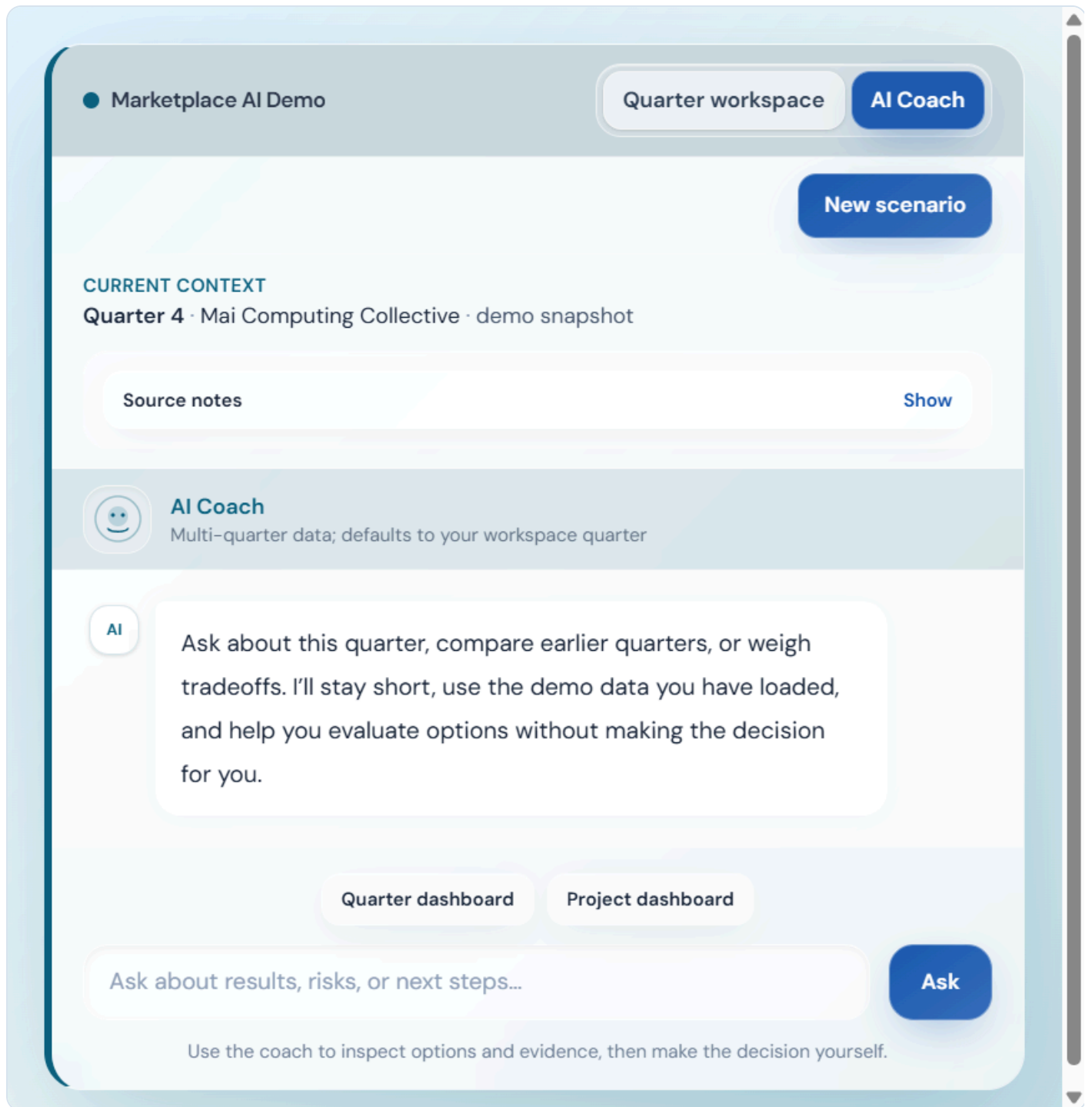
Real teams already use a combination of structured operational data and language models when they need to move quickly on performance reviews, variance stories, competitive moves, capacity and demand risk, and questions about what changed before a planning or leadership meeting. The pattern is consistent even when the vendor stack changes. Dashboards and ERP style systems hold the truth, retrieval or tool calls keep the model tied to that truth, and the model adds short explanations, comparisons, and tradeoff framing so people can reach a better discussion faster without pretending the spreadsheet or report no longer matters.

That is the same muscle as an internal briefing where someone says here is what moved, here is what worries me, and here are two ways to read it, except the system can support the first draft and the follow up questions in real time. The role might sit in finance, operations, or strategy, but the underlying function is still interpretation and preparation, with humans responsible for the final decision. Marketplace Analyst is a student facing version of that practice. It helps the learner see the story in the data instead of outsourcing the strategy itself.

2. How could this type of AI be incorporated into a Marketplace simulation?

Marketplace Analyst is designed to live inside the student workflow between rounds. Instead of sending the student to a separate homework style chatbot, it places a coach beside the same reports they already need to interpret. A student can ask in ordinary language what matters for their team this quarter and receive short, grounded answers tied to the numbers already loaded for that run.

The coach opens with the active quarter and team already in context, which keeps the interaction specific to the student's simulation state rather than generic business advice. It also exposes source notes that feel like a lightweight index of the key facts the assistant can reference from performance, manufacturing, finance, and scorecard views.



AI Coach home with the active quarter, team context, and direct dashboard shortcuts.

When those source notes are expanded, the student can immediately see that the experience is grounded in named facts and report areas rather than vague summaries. That makes the tool feel inspectable and accountable, which matters both pedagogically and practically.

New scenario

CURRENT CONTEXT

Quarter 4 · Mai Computing Collective · demo snapshot

Source notes

Hide

[perf-share](#) · Performance & reports: Team overall share 18.4% after Q3 results; Core stockout, Nomad ending inventory 23 units.

[score-mfg](#) · Balanced scorecard: Manufacturing productivity score 63 (↓ vs 66) — capacity strain / stockout pattern.

[mkt-price](#) · Marketing → Pricing: Core price 2,750; Nomad 3,350; rebates 200 each.

[mfg-cap](#) · Manufacturing: Operating capacity 820 units; utilization 94%; forecast demand Core 455 / Nomad 382 next quarter.

[fin-cash](#) · Finance: Ending cash 589k; dividends 40k; short-term debt 220k.

[hr-prod](#) · Human resources: Sales productivity index 102 vs cohort baseline 100; training 6.5 hrs/quarter.

[graph-appeal](#) · Strategic graphs: Market appeal index rose Q1→Q3 results from 62 to 73; share tracked 12.1% to 18.4%.



AI Coach

Multi-quarter data; defaults to your workspace quarter

AI

Ask about this quarter, compare earlier quarters, or weigh tradeoffs. I'll stay short, use the demo data you have loaded,

Expanded source notes showing linked facts from the loaded quarter snapshot.

From there the student can ask for an interpretation, a risk check, or a visual overview. In the current prototype, one tap can request a quarter dashboard inside the conversation so the reply becomes a compact briefing with revenue, share, net income, and a path to go deeper.

Create a dashboard for this quarter. Include the dashboard_preview widget after a brief intro.

AI

Here's a quick Quarter 4 dashboard view for Mai Computing Collective. Share is 18.4%, revenue is \$2.30M, and net income is \$214.5k [\[Overall share\]](#) [\[Quarter overview\]](#) .

QUARTER

Quarter 4 dashboard

Watch capacity strain and stockout pressure alongside cash and brand results
[mfg-cap][brand-core]

REVENUE

\$2.3M

SHARE

18.4%

NET

\$215k



Open dashboard

Quarter dashboard

Project dashboard

Ask about results, risks, or next steps...

Ask

Use the coach to inspect options and evidence, then make the decision yourself.

Quarter dashboard preview rendered directly inside the conversation.

If the student wants more than a quick summary, the preview opens a richer dashboard layer with overview, market and brands, financial, and operations tabs. This makes the placement inside Marketplace easy to imagine. The assistant is not replacing the workspace. It is adding a guided readout beside the same KPI trail the course already rewards.

Quarter 4 dashboard

Quarter 4 snapshot with performance, brands, capacity, cash, and scorecard themes [perf-share][fin-cash][score-mfg]



Close

Overview

Market & brands

Financials

Operations

REVENUE
\$2.3M

NET INCOME
\$215k

ENDING CASH
\$589k

OVERALL
SHARE
18.4%

SCORECARD
INDEX
74

EXECUTIVE OVERVIEW

Mai Computing Collective — Quarter 4. Strategy: Premium reliability in Workhorse + Traveler; disciplined capacity and margin guardrails. Target segments: Workhorse, Traveler. This quarter closed at **\$2.3M** revenue, **\$215k** net income, and **18.4%** overall share with a cumulative scorecard index of 74.

- Lead profit engine: Core at \$341k brand profit — watch demand (430 units) vs sold (430).

- 1 brand flagged stockout — revisit production and inventory before the next decision round.

Expanded dashboard overlay with KPI summary and executive overview.

Decision lens: this is mainly a choice between protecting the clearest unmet demand signal versus leaning into the higher-priced brand under tight capacity [\[Demand & capacity\]](#) [\[Core performance\]](#) .

Show competitor market share as a chart only after a one sentence intro.

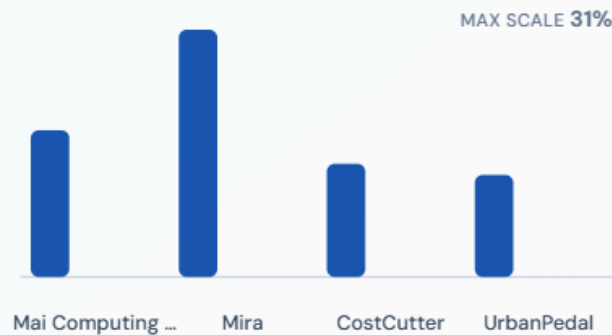
AI

Here's the competitor share view for the active quarter, where your team holds 18.4% and Mira leads at 31% [\[Overall share\]](#) [\[Mira profile\]](#) .

Competitor market share

Current quarter share positioning

● Share %



Active quarter share comparison, including your 18.4% and Mira at 31%

[\[Overall share\]](#)

[\[Mira profile\]](#) .

Example chart widget: competitor market share for the active quarter, rendered inside the coach thread.

quarter forecast is 455 [\[Core performance\]](#) [\[Demand & capacity\]](#). The upside is protecting demand you are already missing. The tradeoff is less room for Nomad, even though total capacity is tight at 820 units with 94% utilization [\[Demand & capacity\]](#). Option 2: Prioritize Nomad Nomad has the higher price at \$3,350 versus Core at \$2,750, and still produced \$319.7k brand profit on lower sold volume [\[Core performance\]](#). The upside is leaning into higher unit revenue. The tradeoff is Nomad ended with 23 units in inventory while Core had zero, so demand risk looks more immediate on Core [\[Core performance\]](#) [\[Overall share\]](#).

Core vs. Nomad next quarter

PRIORITIZE CORE

Defend the brand that already hit a stockout and is forecast at 455 units next quarter [\[Core performance\]](#) [\[Demand & capacity\]](#).

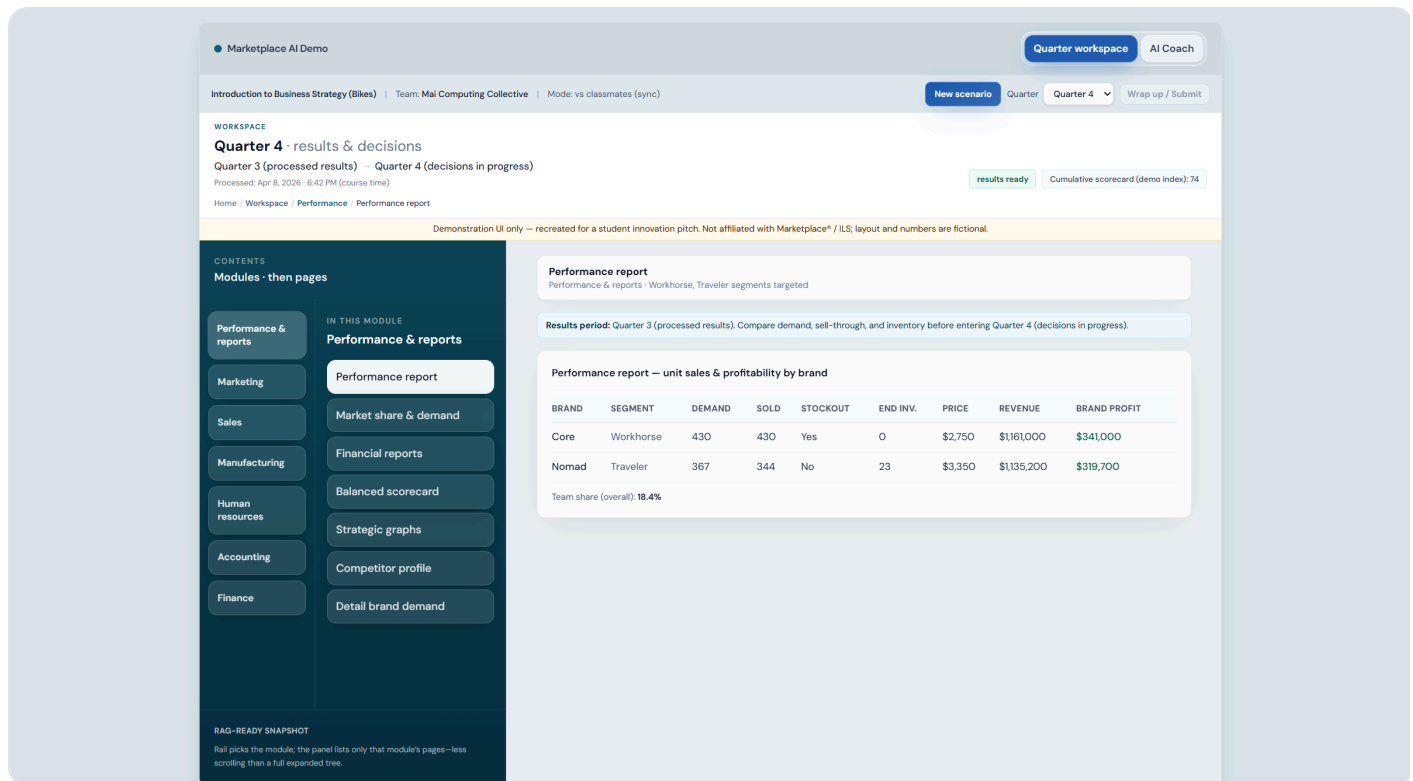
- Core sold 430 and ended at 0 inventory [\[Core performance\]](#)
- Core forecast next quarter: 455 units [\[Demand & capacity\]](#)
- Main risk: tighter capacity for Nomad with utilization already 94% [\[Demand & capacity\]](#)

PRIORITIZE NOMAD

Lean toward the higher-priced brand, which generated \$1.135M revenue and \$319.7k profit this quarter [\[Core](#)

Example tradeoff comparison widget: side by side framing of strategic options with linked citations.

The other half of the experience is the workspace itself. Citations from chat are meant to land in the same module shell students already use, which reinforces that the coach is a voice layer on top of reports they still need to read, question, and defend.



Quarter workspace showing the simulation style report environment the assistant connects back into.

In practice, students would use this after results post, during debrief, and before locking inputs. Across the term it reinforces evidence based reasoning for presentations and team defense in the same way a manager backs a recommendation with numbers in a real meeting.

Live demo reference: The current prototype is publicly available at <https://marketplace-analyst.vercel.app>.

3. What AI tools or applications could be used to support this idea?

The demonstration is built with a standard web stack so it is easy to host, inspect, and extend. The browser talks to one origin that serves the React interface and the API routes. On the server, the live snapshot is compiled into text, lightweight retrieval is run over that text for the user's question, and OpenAI is called through a responses style flow with function tools wired to the same snapshot so KPI pulls are not invented in free text. Vercel runs the static build and the serverless handlers for chat and health while secrets stay off the client. None of that is exotic compared with what a university or vendor team could support in a pilot.

In practical terms, the system performs structured reads and plain language narration. It explains share, cash, manufacturing pressure, and scorecard movement, and when a chart or tradeoff panel is helpful, the interface renders that as a first class visual block instead of hiding the logic inside one long answer. What the student receives is readable language, optional visuals, and links that take them from an answer into the exact part of the workspace that holds the supporting evidence.

For learning, that fits the pattern of workplace analytics copilots tied to ERP or BI systems more closely than a novelty chatbot. The goal is not to print a single best strategy. The goal is to help students ask better questions, interpret the quarter more clearly, and defend their choices with evidence.

Feasibility

The idea is incremental by design. It attaches a coach UI to structured simulation state, adds retrieval and tool backed reads, and ships on standard hosting. That is a reasonable scope compared with building a fully autonomous agent that pretends to run an entire firm. The public demo lets reviewers click through the concept rather than imagine it from a wireframe alone.

Supporting screenshots for this submission are included directly in this PDF and are also available in the project under `assets/r1-proposal-*.png` and `assets/r1-proposal-widget-*.png`. Live demo: <https://marketplace-analyst.vercel.app>.